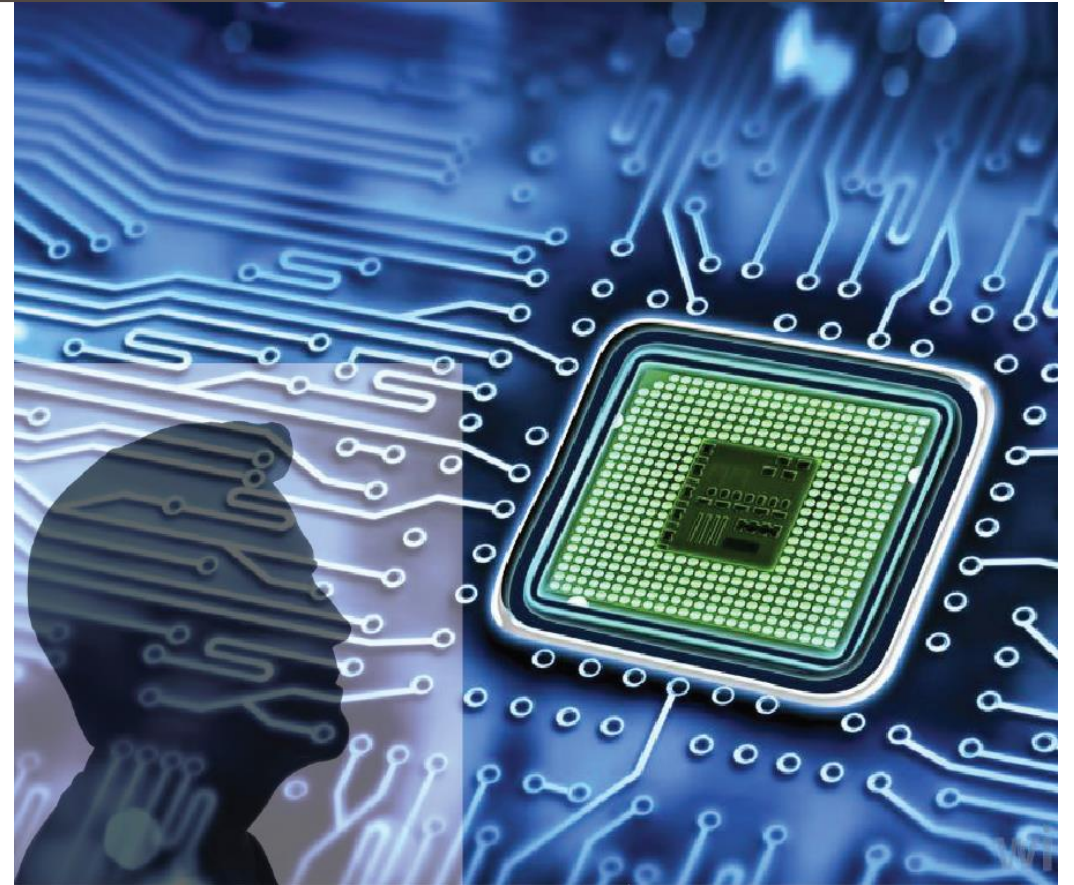


VLSI DESIGN FLOW

Week-13 Lecture-2 Physical Design

Sneh Saurabh
20th November, 2024



VDF: Revision/Announcements

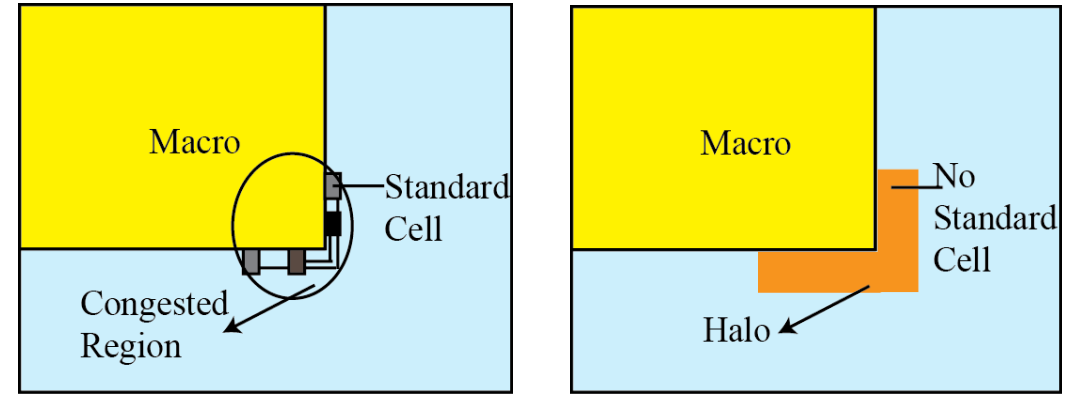
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Floorplanning

Macros Placement: Guidelines (2)

- Add halos around corners
 - Halos are placement blockages



Use flexibility of:

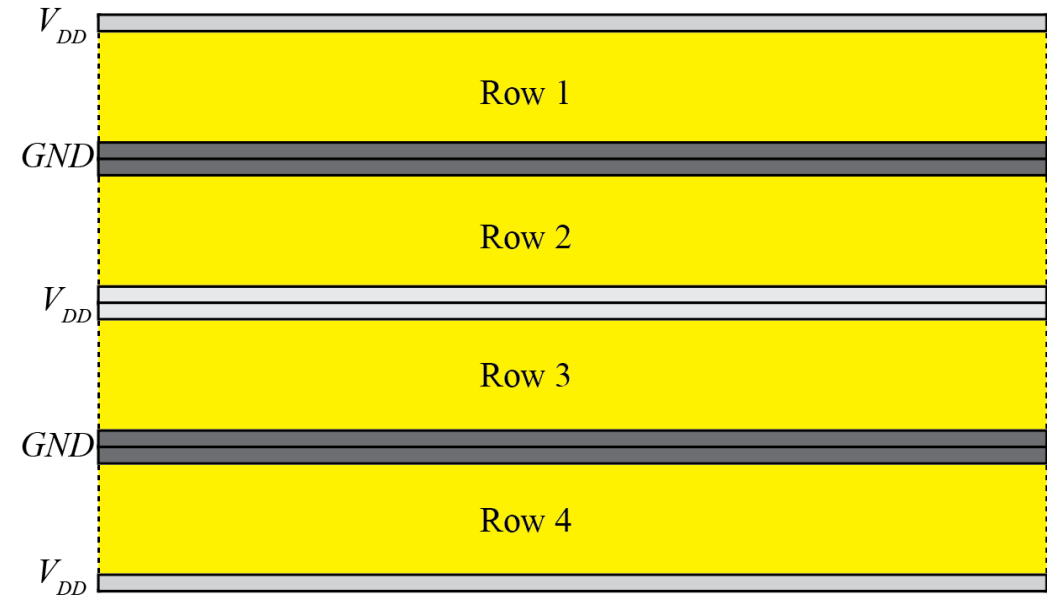
- Orientation: reflection, rotation by 90°
- Shapes: rectilinear
- Pin assignment: locations of movable pins

Floorplanning: Standard Cell Rows

- Height of standard cell rows is equal to the height of the standard cells in the library
 - Multiple rows of different heights can also be created

- Standard cell rows generally created by abutment
- Rows are created oriented in alternating 180-degree rotation

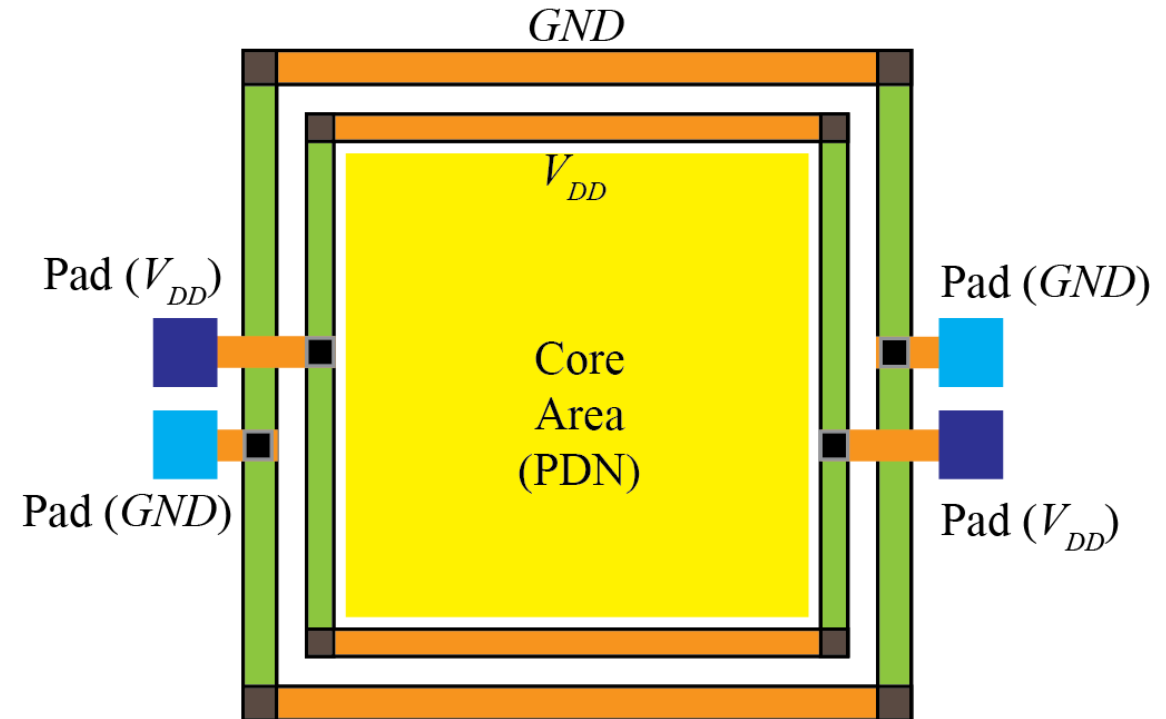
- Routing channels can be created between rows to avoid congestion



Power Planning

Power Planning: Components

- **Power Pads:** Supply power to the chip from the external world
- **Power Rings:** carry power around the periphery of the die
 - Usually use top metal layers
 - Power Delivery Network (PDN) built in the core area



Power Delivery Network (PDN)

Mesh Grid Topology:

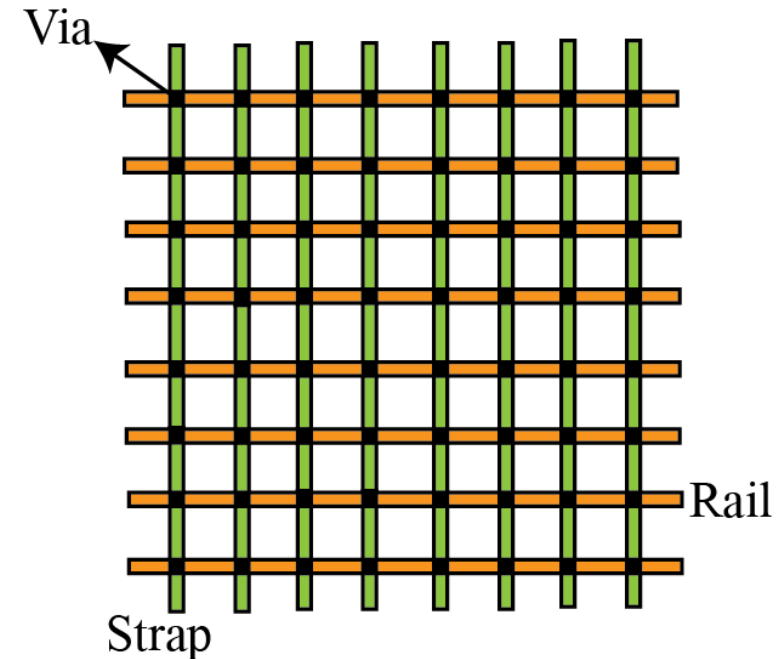
- Popular topology for PDN
- Rails and straps usually organized as a uniformly spaced array

Advantages of mesh grid topology

- Provides low resistance
- Reliability increases
- Uniform current distribution

Design Parameters:

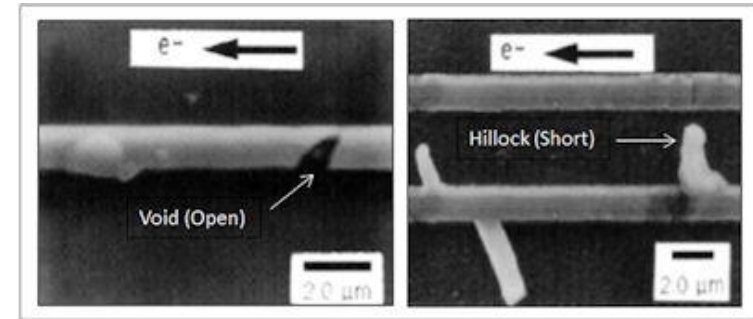
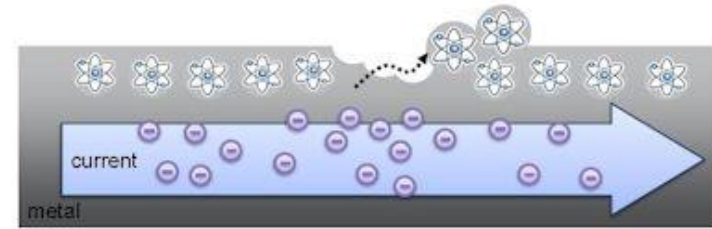
- Layers in which the grid spans
- Width and spacing of wires



Electromigration

Electromigration

- Flow of unidirectional current in a metal can cause **transport of metal mass** in the direction opposite to current flow
 - Ions gain momentum from electrons
 - May eventually lead to shorts and open
-
- PDN prone to electromigration
 - Current in PDN needs to be kept within limit defined by process technology
 - Ensure conductor size is adequate
 - Ensure there are no current density “hot spots”



Source: G. Rangarajan, J. Deng, “Addressing signal electromigration (EM) in today’s complex digital designs”, *EE Times*, Jan. 2013

Voltage Drop

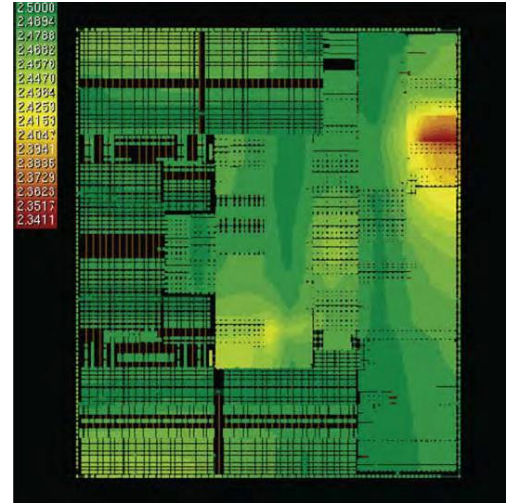
Cause of Voltage Drop:

- Resistance (R): between supply voltage origin and load
- Inductance (L): mainly between package-to-die interconnection
- Static and Dynamic Voltage Drop

$$Voltage = iR + L \frac{di}{dt}$$

Impact on Performance

- Slows down signal propagation along the path
- Maximum frequency at which a circuit will operate is limited
- Need to perform voltage drop analysis in both static and dynamic conditions
 - Fix IR drop hotspots

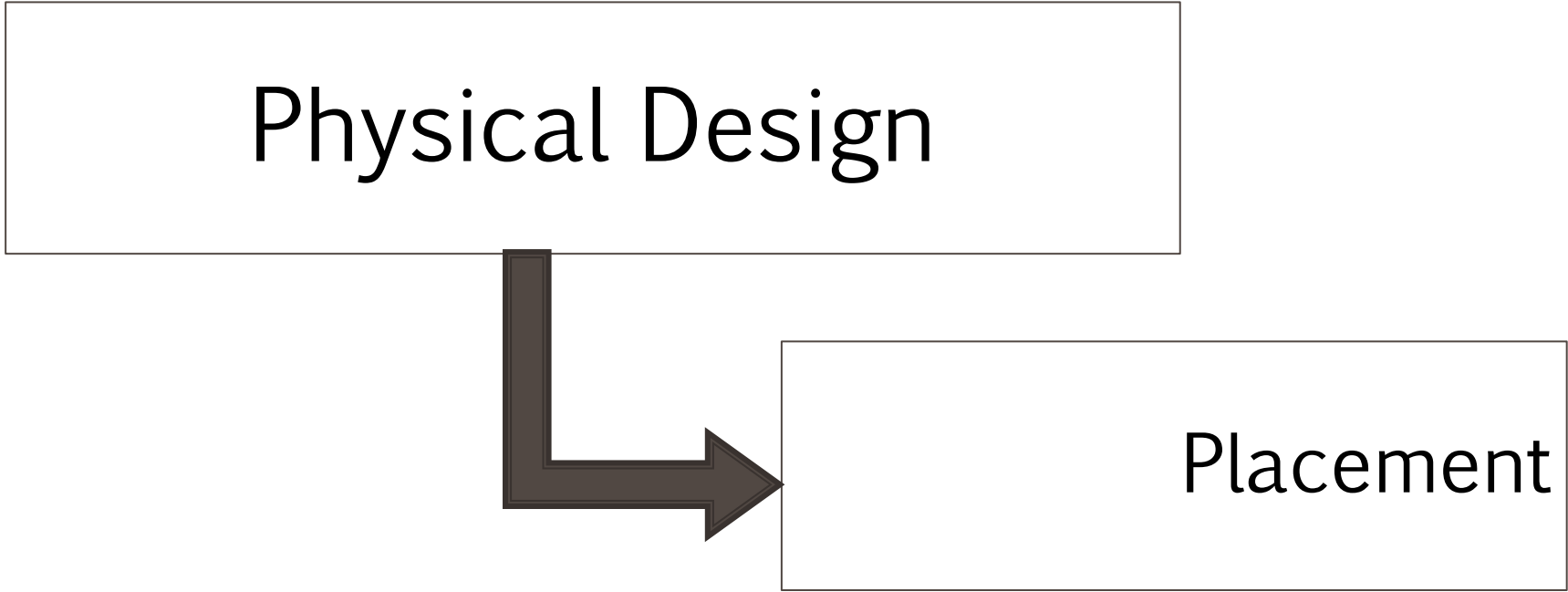


Post-layout IR-drop map of PowerPC™ 750 microprocessor

Source: Electronic Design Automation for Integrated Circuits Handbook, Second Edition, Luciano Lavagno, ...

Decoupling Capacitors (Decap Cells)

- Capacitors inserted between power and ground wires
 - Acts as local charge storage
 - Placed on die at strategic locations
- Demerits: increase die area and leakage power dissipation
- Number of decap cells, size and placement should be optimized

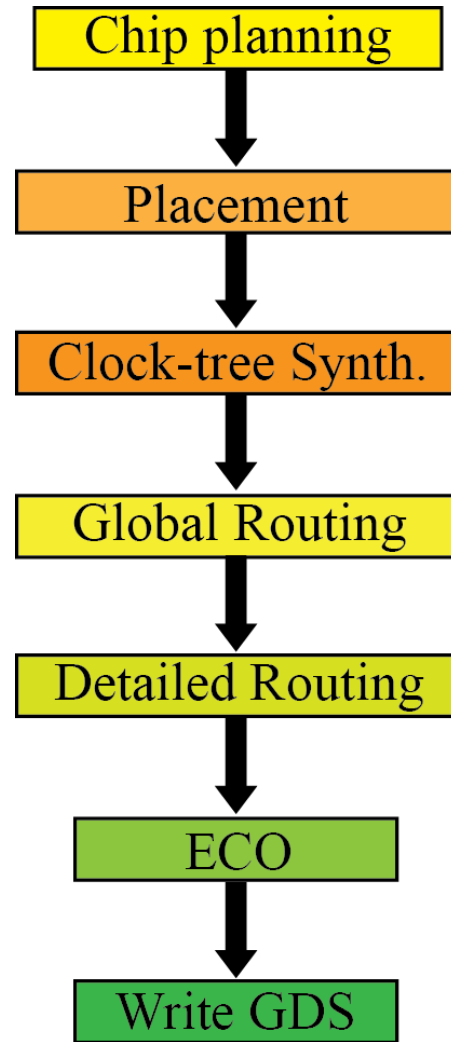


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graph TD; A[Physical Design] --> B[Placement]
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Physical Design

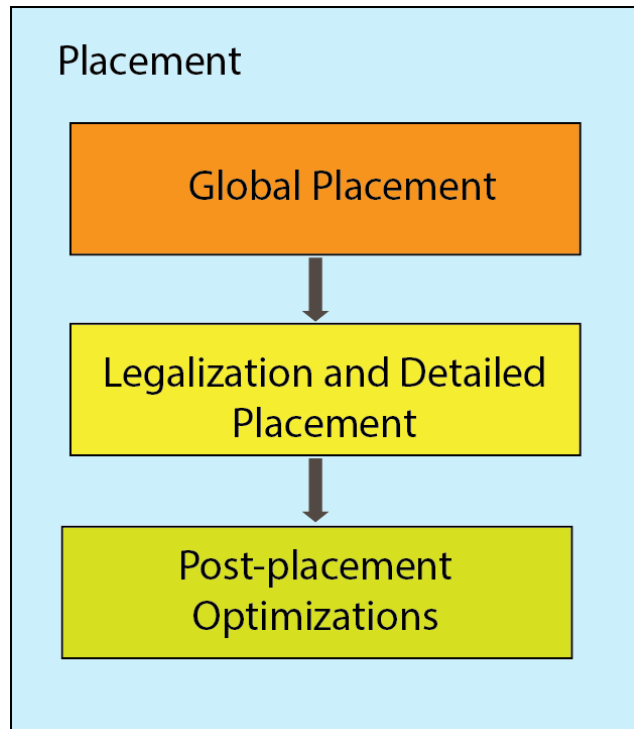
Placement

Physical Design



Placement: Basics

- Decide locations of standard cells
- Goal: to ensure routability of a design
- Metrics: total wirelength
 - Other metrics: timing, congestion



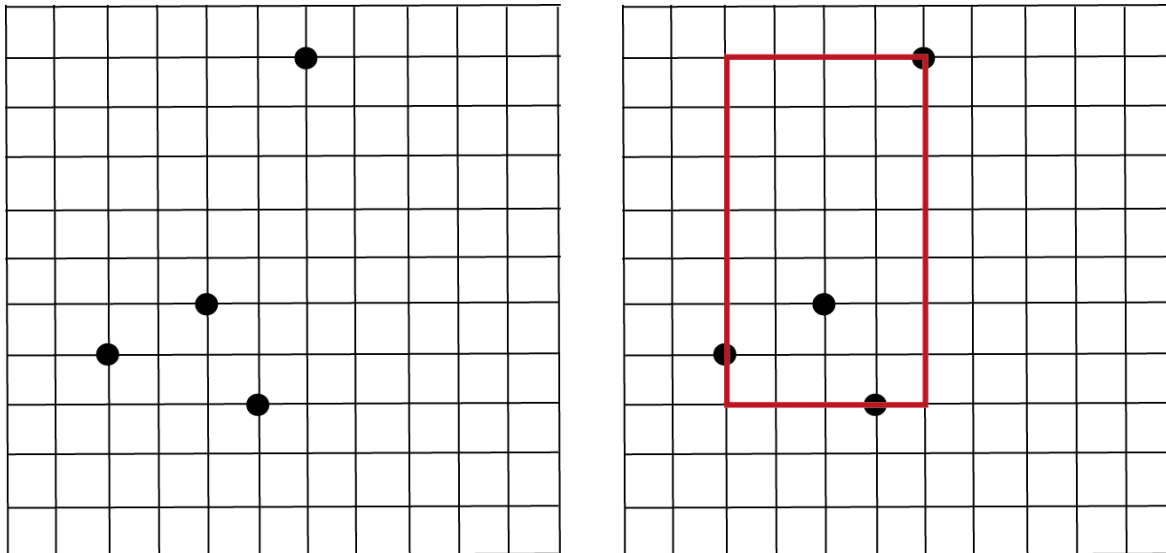
- Global Placement
 - Cells are spread over the core area
 - Overlap may exist
- Legalization and Detailed Placement
 - Overlap between cells removed
 - Cells moved to “legal” positions (legalization)
- Post-Placement Optimizations
 - Buffering, resizing, etc.

Placement: Wirelength Estimates

- Placer needs to compute the wirelength of each net multiple times
- Ideally, estimated wirelength should match post-routing wirelength

Half-perimeter Wirelength (HPWL)

- Easy to compute
- Half of the perimeter of the bounding rectangle that encloses all the pins of a net
- Widely used



Half-perimeter wirelength=11

Global Placement Techniques

- Well-studied problem: since 1970s
- Various heuristics have been proposed

Analytical Placement Algorithm:

- Considers each cell as point object with co-ordinates (x_i, y_i)
- Cost function (such as wirelength) and constraints are defined mathematically
 - Wirelength a function of $(x_i - x_j)^2 + (y_i - y_j)^2$
- Use efficient solvers to obtain minimum cost for the mathematical formulations
 - Suitable constraints of fixed entities
 - Cost related to cell density added
- Cells allowed to overlap in global placement
 - Legalization becomes necessary

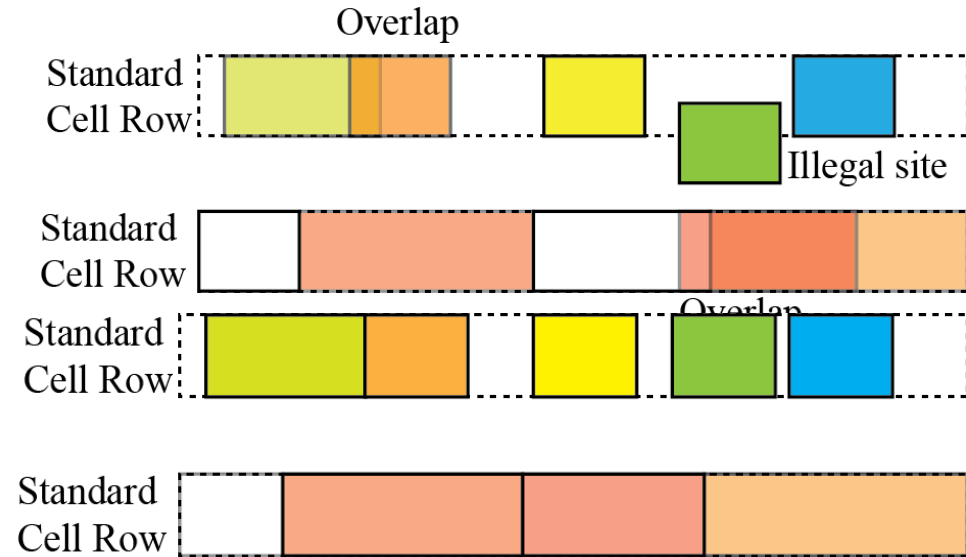
Placement: Legalization and Detailed Placement

A placement is illegal if:

- Cells overlap
- Cells occupy illegal sites (for example between placement rows)

Legalization:

- Removes all overlaps and snap cells to legal sites
 - With the minimum impact on the wirelength, timing and congestion



Detailed placement:

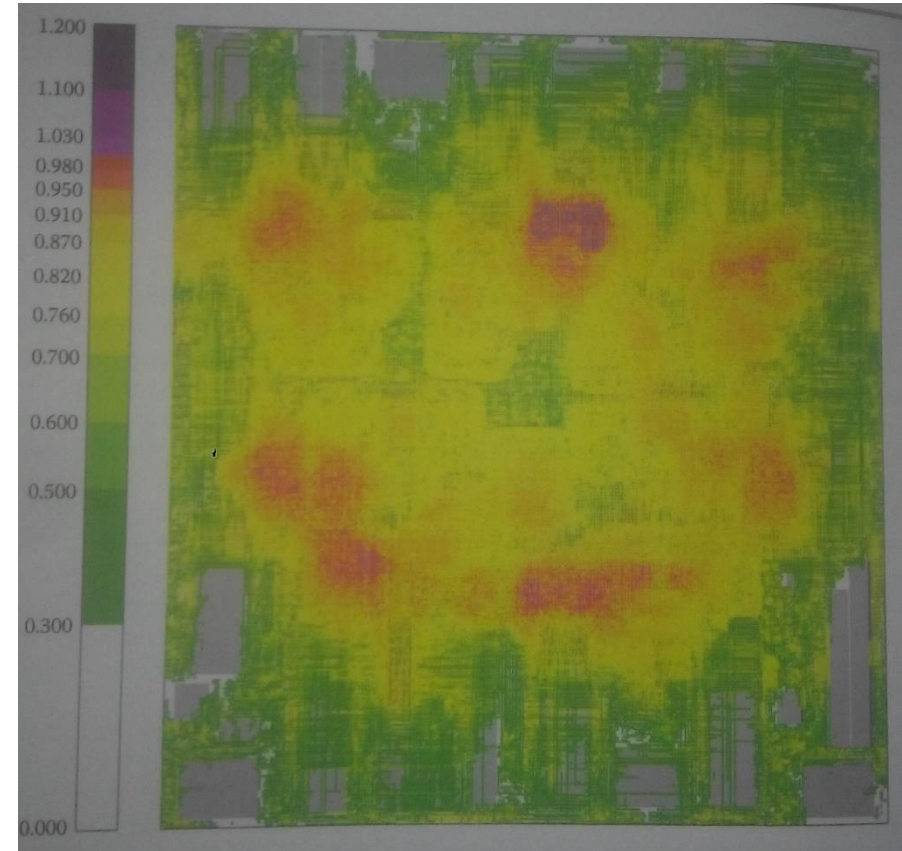
- Improves the QoR by incremental changes to the cell location(s)
- Improves wirelength and routability by:
 - Swapping location of neighboring cells
 - Re-distributing free sites
 - Moving cells to unused location

Timing-driven Placement

- Perform timing analysis internally and incrementally during placement
 - Target Worst Negative Slack (WNS) and Total Negative Slack (TNS)
- Control the proximity of cells that are on critical paths
- Depending on placement algorithm, different approaches to obtain timing driven placement is taken
- Additional weight is added to nets to indicate timing criticality
 - Give more weights to nets that are timing critical
 - Additional weights bias the placement engine to place timing critical objects together

Placement: Congestion Driven

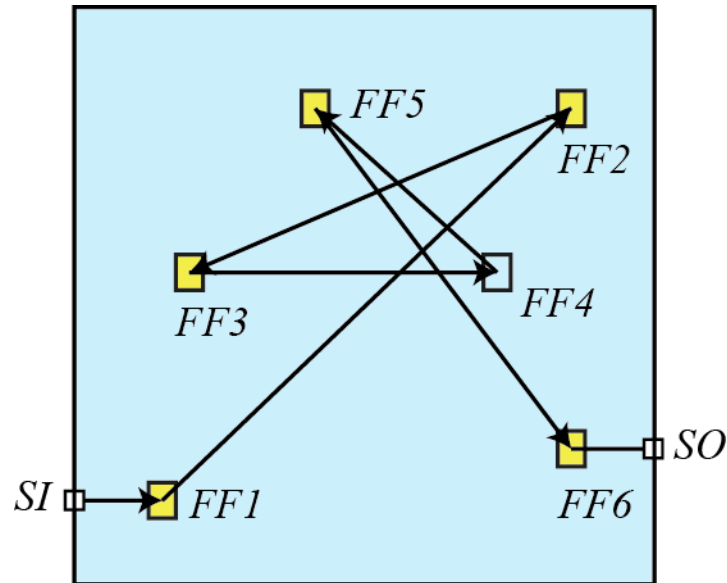
- **Congestion:** Routing Resource Demand more than Routing Resource Supply
- Congestion leads to:
 - Failure in routing
 - Detour: increase in wirelength
- Divide layout in rectangular regions
- Carry out fast global routing
- Identify *congestion hot spots*
- Move cells out of region
- Expand region for placement



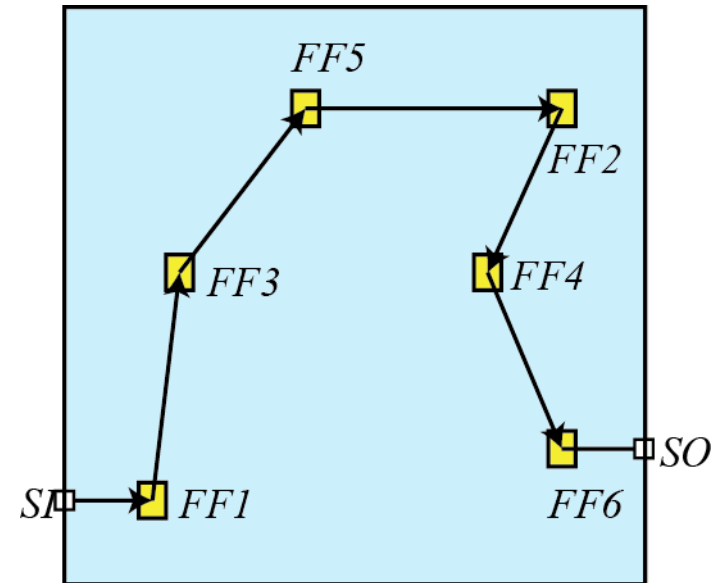
Source: Electronic Design Automation for Integrated Circuits Handbook, Second Edition, Luciano Lavagno, Igor L. Markov, Grant E. Martin, Louis K. Scheffer, - 2016, - CRC Press

Scan Chain Reordering

- Scan cells that are nearby form consecutive flip-flops in the scan chain



Before Reordering



After Reordering

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.